

```
<< FeynCalc`  
Context[ChiralityProjector]  
Names["*"ChiralityProjector"]  
$Packages  
$FeynCalcVersion
```

FeynCalc 10.1.0 (stable version). For help, use the [online documentation](#), visit the [forum](#) and have a look at the supplied [examples](#). The PDF-version of the manual can be downloaded [here](#).

If you use FeynCalc in your research, please
evaluate FeynCalcHowToCite[] to learn how to cite this software.

*Please keep in mind that the proper academic attribution
of our work is crucial to ensure the future development of this package!*

FeynCalc`

{ChiralityProjector}

```

(FeynCalc`, Parallel`Protected`, Parallel`Developer`,
Parallel`Evaluate`, Parallel`Combine`, Parallel`Queue`Priority`,
Parallel`Queue`FIFO`, Parallel`Queue`Interface`, Parallel`Concurrency`,
Parallel`VirtualShared`, Parallel`Status`, Parallel`Settings`,
Parallel`Parallel`, Forms`, KernelObjects`KernelEditor`,
Evaluators`WSTPServerKernels`, ProcessLink`, Evaluators`SshKernels`,
Evaluators`SelfKernels`, Evaluators`LWGKernels`, Evaluators`LocalKernels`,
Evaluators`CloudKernels`, KernelObjects`Implementations`, Security`,
Interpreter`, KernelConfiguration`, KernelObject`, Parallel`Kernels`,
Parallel`, Parallel`Debug`Perfmon`, Parallel`Debug`, ErrorHandling`,
TypeSystem`, Macros`, DocumentationSearch`Skeletonizer`, JLink`,
DocumentationSearch`, GetFEKernelInit`, RemoteEvaluate`, KeychainLink`,
GeneralUtilities`, EntityFramework`WolframAlphaLastUpdate`,
EntityFramework`WolframAlphaEntityStore`,
EntityFramework`WolframAlphaDataUsingCache`, EntityFramework`WolframAlphaData`,
EntityFramework`Utilities`, EntityFramework`ToFromEntity`,
EntityFramework`SideEffects`, EntityFramework`Registry`,
EntityFramework`RandomEntity`, EntityFramework`PurgeEntityCache`,
EntityFramework`Preferences`, EntityFramework`Predicates`,
EntityFramework`PopulateEntityCache`, EntityFramework`OperatorForms`,
EntityFramework`InterpreterCache`, EntityFramework`Formatting`,
EntityFramework`Filter`, EntityFramework`FileEntityCache`,
EntityFramework`FileAssociation`, EntityFramework`EvaluateWrapper`,
EntityFramework`EvaluateQualified`, EntityFramework`EvaluateExists`,
EntityFramework`EvaluateEntity`, EntityFramework`EvaluateComplexEntity`,
EntityFramework`EntityValue`, EntityFramework`EntityTypeValue`,
EntityFramework`EntityType`, EntityFramework`EntityStoreFeatures`,
EntityFramework`EntityStoreBoxes`, EntityFramework`EntityStore`,
EntityFramework`EntityProperties`, EntityFramework`EntityPrefetch`,
EntityFramework`EntityMetaInformation`, EntityFramework`EntityList`,
EntityFramework`EntityInstance`, EntityFramework`EntityFunction`,
EntityFramework`EntityDuration`, EntityFramework`DisambiguateEntityProperty`,
EntityFramework`DefaultEntityTypes`, EntityFramework`DataUtilities`,
EntityFramework`DataFunctions`, EntityFramework`CustomEntity`,
EntityFramework`CreateEntityPrefetchArchive`,
EntityFramework`ComputedProperties`, EntityFramework`CommonName`,
EntityFramework`CanonicalName`, EntityFramework`Caching`, SQLiteLink`,
EntityFramework`CachedEntityStoreValue`, EntityFramework`BatchMap`,
EntityFramework`BatchDownload`, EntityFramework`BatchApplied`, EntityFramework`,
QuantityUnits`, CURLLink`URLResponseTime`, CURLLink`Utilities`, CURLLink`Info`,
CURLLink`Cookies`, CURLLink`HTTP`, OAuthSigning`, CURLLink`URLFetch`, CURLLink`,
URLUtilities`, Templating`, Wolfram`Chatbook`, CloudObjectLoader`, UnitTable`,
CloudExpressionLoader`, ResourceFunctionHelpers`, Wolfram`LLMFunctions`,
AstronomyConvenienceFunctionsLoader`, AR`Load`, Polygons`Load`, CSGRegion`Load`,
Regions`Load`, NurbsRegion`Load`, ImportExport`Load`, WebAudioSearchLoader`,
NeuralNetworks`, NeuralFunctions`, NaturalLanguageProcessingLoader`,
NaturalLanguage`, MobileMessaging`, MachineLearningLoader`, IntegratedServices`,
IconizeLoader`, HTTPHandling`, GeoTools`, GeometrySearchLoader`, FormNotebook`,
ExternalEvaluate`, DatasetLoader`, Cryptography`, CloudDialogs`, Blockchain`,
Authentication`, SystemTools`, SPICELink`, SecureShellLink`, ProcessLinkLoader`,
MailLink`, IMAPLink`, ExternalStorage`, QuestionFramework`, ToolbarSupport`,
WolframAppCatalog`, ResourceLocator`, PacletManager`, PersistenceLocations`,
System`, Global`, Wolfram`Chatbook`Actions`, Wolfram`Chatbook`ChatGroups`,
Wolfram`Chatbook`ChatHistory`, Wolfram`Chatbook`ChatMessages`,
Wolfram`Chatbook`ChatModes`, Wolfram`Chatbook`ChatModes`Common`,
Wolfram`Chatbook`ChatModes`Context`, Wolfram`Chatbook`ChatModes`Evaluate`,
Wolfram`Chatbook`ChatModes`ShowCodeAssistance`, Wolfram`Chatbook`ChatModes`UI`,
Wolfram`Chatbook`ChatState`, Wolfram`Chatbook`CloudToolbar`,
Wolfram`Chatbook`Common`, Wolfram`Chatbook`CreateChatNotebook`,
Wolfram`Chatbook`Dialogs`, Wolfram`Chatbook`Dynamics`,
Wolfram`Chatbook`Errors`, Wolfram`Chatbook`ErrorUtils`,
Wolfram`Chatbook`Explode`, Wolfram`Chatbook`Feedback`,
Wolfram`Chatbook`Formatting`, Wolfram`Chatbook`FrontEnd`,
Wolfram`Chatbook`Handlers`, Wolfram`Chatbook`InlineReferences`,
Wolfram`Chatbook`Menus`, Wolfram`Chatbook`Models`,
Wolfram`Chatbook`PersonaManager`, Wolfram`Chatbook`Personas`,
Wolfram`Chatbook`PreferencesContent`, Wolfram`Chatbook`PreferencesUtils`,
Wolfram`Chatbook`Prompting`, Wolfram`Chatbook`ResourceInstaller`,
Wolfram`Chatbook`Sandbox`, Wolfram`Chatbook`SendChat`,
Wolfram`Chatbook`Serialization`, Wolfram`Chatbook`Services`,
Wolfram`Chatbook`Settings`, Wolfram`Chatbook`ToolManager`,
Wolfram`Chatbook`Tools`, Wolfram`Chatbook`UI`, Wolfram`Chatbook`Utils`)

```

```

ClearAll["Global`*"];
FCClearScalarProducts[];

SP[P1, P1] = Ms^2;
SP[P2, P2] = Ms^2;
SP[k1, k1] = 0;
SP[k2, k2] = m1^2;

qMom = P1 - P2;
Lcur[mu_] := GA[mu] . (1 - GA[5]);

Vtx[mu_] := CV GA[mu] - I CM / (2 MN) DiracSigma[GA[mu], GS[qMom]] - CA (GA[mu] . GA[5] - Fq FV[qMom, mu] × GA[5]);

VtxBar[nu_] := CV GA[nu] + I CM / (2 MN) DiracSigma[GA[nu], GS[qMom]] - CA (GA[nu] . GA[5] + Fq FV[qMom, nu] × GA[5]);

ds[expr_] := DiracSimplify[expr, ToDiracGamma67 → False, DiracTraceEvaluate → True];

Ltr[mu_, nu_] := ds[DiracTrace[(GS[k2] + m1) . Lcur[mu] . GS[-k1] . Lcur[nu]]];

Htr[mu_, nu_] :=
ds[
  DiracTrace[
    (GS[P2] + Ms)
    . Vtx[mu]
    . (GS[P1] + Ms)
    . VtxBar[nu]
  ]
];

MsqFC =
- GF^2 Ccab^2 / 2 Contract[Ltr[mu, nu] × Htr[mu, nu]] //
FCI //
Contract //
ExpandScalarProduct //
ds //
ComplexExpand //
Together //
FullSimplify //
Factor;

s[a_, b_] := SP[a, b];

eq41Bracket = (CA^2 - CV^2) Ms^2 s[k1, k2] + (CA - CV)^2 s[k1, P2] × s[k2, P1] + (CA + CV)^2 s[k1, P1] × s[k2, P2] +
CM Ms / MN ( 2 CA ( s[k1, P1] × s[k2, P2] - s[k1, P2] × s[k2, P1] ) + CV ( s[k1, k2] (s[P1, P2] - Ms^2) - (s[k1, P1] - s
CM^2 / (4 MN^2) ( Ms^2 s[k1, P2] (3 s[k2, P2] - s[k2, P1]) + Ms^2 s[k1, P1] (3 s[k2, P1] - s[k2, P2]) + s[k1, k2] (s[P1, P2] -
CA^2 Fq ( 2 Ms + Fq (Ms^2 - s[P1, P2]) ) ( s[k1, k2] (Ms^2 - s[P1, P2]) - (s[k1, P1] - s[k1, P2]) (s[k2, P1] - s[k2, P2])

Eq41 = 32 GF^2 Ccab^2 eq41Bracket // FCI // Contract // ExpandScalarProduct // FullSimplify // Factor;
difference =
FullSimplify[MsqFC - Eq41 //
FCI //
Contract //
ExpandScalarProduct //
DiracSimplify[#, ToDiracGamma67 → False] & //
ComplexExpand //
Together //
FullSimplify //
Factor]

```

```

ClearAll[mu, nu, P1, P2, k1, k2, qMom, Q2, Ms1, Ms2, M1, M2, Ms, MN, ml, GF, Ccab, CV, CA, CM, Fq, F1V, F2V, F3V, F1A, F2A, F3A,
Eq41GeneralBracket, Eq41GeneralShort, Eq41GeneralCollected, s, OldT1, OldT2, OldT3, OldT4, OldT5, OldT6, Eq41OldBracket, Eq41O

FCClearScalarProducts[];

qMom = P1 - P2;

Q2 = SP[qMom, qMom];

SP[P1, P1] = Ms1^2;

SP[P2, P2] = Ms2^2;

SP[k1, k1] = 0;

SP[k2, k2] = ml^2;

G5 := GA[5];

Lcur[mu_] := GA[mu].(1 - GA[5]);

Vtx[mu_] := F1V[Q2] * GA[mu] - F1A[Q2] * GA[mu].G5 + I F2V[Q2] * DiracSigma[GA[mu], GS[qMom]] / (M1 + M2) - I F2A[Q2] * DiracSigma[GA[mu], C
ClearAll[DiracAdjointFC];

DiracAdjointFC[x_Plus] := DiracAdjointFC /@ x;

DiracAdjointFC[x_Times] := Times @@ (DiracAdjointFC /@ (List @@ x));

DiracAdjointFC[x_Dot] := Dot @@ Reverse[DiracAdjointFC /@ (List @@ x)];

DiracAdjointFC[GA[5]] := -GA[5];

DiracAdjointFC[GA[a_]] := GA[a];

DiracAdjointFC[GS[p_]] := GS[p];

DiracAdjointFC[DiracSigma[a_, b_]] := DiracSigma[a, b];

DiracAdjointFC[x_?NumericQ] := Conjugate[x];

DiracAdjointFC[x_] := x;
VtxBar[nu_] := DiracAdjointFC[Vtx[nu]] // DiracSimplify // Contract; (*GA[0].Vtx[nu].GA[0];*)
(*F1V[Q2] GA[nu] + F1A[Q2] GA[nu].G5 + I F2V[Q2] DiracSigma[GA[nu], GS[qMom]] / M1 - I F2A[Q2] DiracSigma[GA[nu], GS[qMom]].G5 / M1 +
(*VtxBar[nu_] := F1V[Q2] GA[nu] - I F2V[Q2] / (M1 + M2) * DiracSigma[GA[nu], GS[qMom]] + F3V[Q2] / M1 * FV[qMom, nu] - F1A[Q2] GA[nu].GA[5] - I F2A[Q2] /
Ltr[mu_, nu_] := DiracTrace[(GS[k2] + ml).Lcur[mu].GS[k1].Lcur[nu], DiracTraceEvaluate -> True] // DiracSimplify // Contract;

Htr[mu_, nu_] := DiracTrace[(GS[P2] + Ms2).Vtx[mu].(GS[P1] + Ms1).VtxBar[nu], DiracTraceEvaluate -> True] // DiracSimplify // Contr

MsqGeneral = GF^2 Ccab^2 Contract[Ltr[mu, nu] * Htr[mu, nu]] / 2 // FCI // Contract // ExpandScalarProduct // DiracSimplify // Comple

Eq41GeneralBracket = MsqGeneral / (32 GF^2 Ccab^2) // Contract // ExpandScalarProduct // DiracSimplify // ComplexExpand // Factor;

Eq41GeneralShort = Eq41GeneralBracket /. {F1V[_] -> f1v, F2V[_] -> f2v, F3V[_] -> f3v, F1A[_] -> f1a, F2A[_] -> f2a, F3A[_] -> f3a};

Eq41GeneralCollected = Collect[Eq41GeneralShort, {f1v, f2v, f3v, f1a, f2a, f3a}, Factor];

FullSimplify[Eq41GeneralCollected]

```

$$\begin{aligned}
& \frac{1}{2} \left(2 \left(\text{Ms1 Ms2} (\overline{\text{k1}} \cdot \overline{\text{k2}}) + (\overline{\text{k1}} \cdot \overline{\text{P2}}) (\overline{\text{k2}} \cdot \overline{\text{P1}}) + (\overline{\text{k1}} \cdot \overline{\text{P1}}) (\overline{\text{k2}} \cdot \overline{\text{P2}}) \right) \text{f1a}^2 + \right. \\
& \quad \frac{4 \text{f2v} (\text{Ms1} + \text{Ms2}) ((\overline{\text{k1}} \cdot \overline{\text{P2}}) (\overline{\text{k2}} \cdot \overline{\text{P1}}) - (\overline{\text{k1}} \cdot \overline{\text{P1}}) (\overline{\text{k2}} \cdot \overline{\text{P2}})) \text{f1a}}{\text{M1} + \text{M2}} + \\
& \quad 4 \text{f1v} ((\overline{\text{k1}} \cdot \overline{\text{P1}}) (\overline{\text{k2}} \cdot \overline{\text{P2}}) - (\overline{\text{k1}} \cdot \overline{\text{P2}}) (\overline{\text{k2}} \cdot \overline{\text{P1}})) \text{f1a} - \\
& \quad \frac{1}{\text{M1}} 2 \text{f3a} ((\overline{\text{k1}} \cdot \overline{\text{P2}}) ((\text{Ms1} + \text{Ms2}) (\overline{\text{k2}} \cdot \overline{\text{P1}}) - 2 \text{Ms1} (\overline{\text{k2}} \cdot \overline{\text{P2}})) + (\overline{\text{k1}} \cdot \overline{\text{P1}}) \\
& \quad \quad ((\text{Ms1} + \text{Ms2}) (\overline{\text{k2}} \cdot \overline{\text{P2}}) - 2 \text{Ms2} (\overline{\text{k2}} \cdot \overline{\text{P1}})) + (\text{Ms1} + \text{Ms2}) (\overline{\text{k1}} \cdot \overline{\text{k2}}) (\text{Ms1 Ms2} - \overline{\text{P1}} \cdot \overline{\text{P2}})) \text{f1a} + \\
& \quad \frac{1}{\text{M1} + \text{M2}} 2 \text{f2a} ((\overline{\text{k1}} \cdot \overline{\text{P2}}) ((\text{Ms1} - \text{Ms2}) (\overline{\text{k2}} \cdot \overline{\text{P1}}) - 2 \text{Ms1} (\overline{\text{k2}} \cdot \overline{\text{P2}})) + (\overline{\text{k1}} \cdot \overline{\text{P1}}) \\
& \quad \quad (2 \text{Ms2} (\overline{\text{k2}} \cdot \overline{\text{P1}}) + (\text{Ms1} - \text{Ms2}) (\overline{\text{k2}} \cdot \overline{\text{P2}})) + (\text{Ms1} - \text{Ms2}) (\overline{\text{k1}} \cdot \overline{\text{k2}}) (\text{Ms1 Ms2} + \overline{\text{P1}} \cdot \overline{\text{P2}})) \text{f1a} - \\
& \quad \frac{4 \text{f1v f2a} (\text{Ms1} - \text{Ms2}) ((\overline{\text{k1}} \cdot \overline{\text{P2}}) (\overline{\text{k2}} \cdot \overline{\text{P1}}) - (\overline{\text{k1}} \cdot \overline{\text{P1}}) (\overline{\text{k2}} \cdot \overline{\text{P2}}))}{\text{M1} + \text{M2}} + \\
& \quad \frac{4 \text{f2a f2v} (\text{Ms1} - \text{Ms2}) (\text{Ms1} + \text{Ms2}) ((\overline{\text{k1}} \cdot \overline{\text{P2}}) (\overline{\text{k2}} \cdot \overline{\text{P1}}) - (\overline{\text{k1}} \cdot \overline{\text{P1}}) (\overline{\text{k2}} \cdot \overline{\text{P2}}))}{(\text{M1} + \text{M2})^2} + \\
& \quad 2 \text{f1v}^2 (-\text{Ms1 Ms2} (\overline{\text{k1}} \cdot \overline{\text{k2}}) + (\overline{\text{k1}} \cdot \overline{\text{P2}}) (\overline{\text{k2}} \cdot \overline{\text{P1}}) + (\overline{\text{k1}} \cdot \overline{\text{P1}}) (\overline{\text{k2}} \cdot \overline{\text{P2}})) + \\
& \quad \frac{1}{\text{M1} + \text{M2}} 2 \text{f1v f2v} ((\overline{\text{k1}} \cdot \overline{\text{P2}}) (2 \text{Ms1} (\overline{\text{k2}} \cdot \overline{\text{P2}}) - (\text{Ms1} + \text{Ms2}) (\overline{\text{k2}} \cdot \overline{\text{P1}})) + \\
& \quad \quad (\overline{\text{k1}} \cdot \overline{\text{P1}}) (2 \text{Ms2} (\overline{\text{k2}} \cdot \overline{\text{P1}}) - (\text{Ms1} + \text{Ms2}) (\overline{\text{k2}} \cdot \overline{\text{P2}})) + (\text{Ms1} + \text{Ms2}) (\overline{\text{k1}} \cdot \overline{\text{k2}}) (\text{Ms1 Ms2} - \overline{\text{P1}} \cdot \overline{\text{P2}})) + \\
& \quad \frac{\text{f3a}^2 ((\overline{\text{k1}} \cdot \overline{\text{k2}}) (\text{Ms1}^2 + \text{Ms2}^2 - 2 (\overline{\text{P1}} \cdot \overline{\text{P2}})) - 2 (\overline{\text{k1}} \cdot \overline{\text{P1}} - \overline{\text{k1}} \cdot \overline{\text{P2}}) (\overline{\text{k2}} \cdot \overline{\text{P1}} - \overline{\text{k2}} \cdot \overline{\text{P2}})) (\text{Ms1 Ms2} - \overline{\text{P1}} \cdot \overline{\text{P2}})}{\text{M1}^2} - \\
& \quad \frac{\text{f3v}^2 ((\overline{\text{k1}} \cdot \overline{\text{k2}}) (\text{Ms1}^2 + \text{Ms2}^2 - 2 (\overline{\text{P1}} \cdot \overline{\text{P2}})) - 2 (\overline{\text{k1}} \cdot \overline{\text{P1}} - \overline{\text{k1}} \cdot \overline{\text{P2}}) (\overline{\text{k2}} \cdot \overline{\text{P1}} - \overline{\text{k2}} \cdot \overline{\text{P2}})) (\text{Ms1 Ms2} + \overline{\text{P1}} \cdot \overline{\text{P2}})}{\text{M1}^2} - \\
& \quad \frac{1}{\text{M1}} 2 \text{f1v f3v} ((\overline{\text{k1}} \cdot \overline{\text{P2}}) ((\text{Ms2} - \text{Ms1}) (\overline{\text{k2}} \cdot \overline{\text{P1}}) + 2 \text{Ms1} (\overline{\text{k2}} \cdot \overline{\text{P2}})) + \\
& \quad \quad (\overline{\text{k1}} \cdot \overline{\text{P1}}) ((\text{Ms2} - \text{Ms1}) (\overline{\text{k2}} \cdot \overline{\text{P2}}) - 2 \text{Ms2} (\overline{\text{k2}} \cdot \overline{\text{P1}})) + (\text{Ms1} - \text{Ms2}) (\overline{\text{k1}} \cdot \overline{\text{k2}}) (\text{Ms1 Ms2} + \overline{\text{P1}} \cdot \overline{\text{P2}})) + \\
& \quad \frac{1}{\text{M1} (\text{M1} + \text{M2})} 2 \text{f2a f3a} ((\overline{\text{k1}} \cdot \overline{\text{P2}}) ((\text{Ms2}^2 - \text{Ms1}^2) (\overline{\text{k2}} \cdot \overline{\text{P1}}) + 2 (\overline{\text{k2}} \cdot \overline{\text{P2}}) (\text{Ms1}^2 - \overline{\text{P1}} \cdot \overline{\text{P2}})) + \\
& \quad \quad (\overline{\text{k1}} \cdot \overline{\text{P1}}) ((\text{Ms2}^2 - \text{Ms1}^2) (\overline{\text{k2}} \cdot \overline{\text{P2}}) + 2 (\overline{\text{k2}} \cdot \overline{\text{P1}}) (\overline{\text{P1}} \cdot \overline{\text{P2}} - \text{Ms2}^2))) + \\
& \quad \frac{1}{\text{M1} (\text{M1} + \text{M2})} 2 \text{f2v f3v} ((\overline{\text{k1}} \cdot \overline{\text{P2}}) ((\text{Ms2}^2 - \text{Ms1}^2) (\overline{\text{k2}} \cdot \overline{\text{P1}}) + 2 (\overline{\text{k2}} \cdot \overline{\text{P2}}) (\text{Ms1}^2 - \overline{\text{P1}} \cdot \overline{\text{P2}})) + \\
& \quad \quad (\overline{\text{k1}} \cdot \overline{\text{P1}}) ((\text{Ms2}^2 - \text{Ms1}^2) (\overline{\text{k2}} \cdot \overline{\text{P2}}) + 2 (\overline{\text{k2}} \cdot \overline{\text{P1}}) (\overline{\text{P1}} \cdot \overline{\text{P2}} - \text{Ms2}^2))) + \\
& \quad \frac{1}{(\text{M1} + \text{M2})^2} \text{f2a}^2 ((\overline{\text{k1}} \cdot \overline{\text{k2}}) (\text{Ms1}^2 + \text{Ms2}^2 - 2 (\overline{\text{P1}} \cdot \overline{\text{P2}})) (\text{Ms1 Ms2} + \overline{\text{P1}} \cdot \overline{\text{P2}}) + \\
& \quad \quad 2 (\overline{\text{k1}} \cdot \overline{\text{P1}}) ((\overline{\text{k2}} \cdot \overline{\text{P2}}) (\overline{\text{P1}} \cdot \overline{\text{P2}} - \text{Ms1 Ms2}) + (\overline{\text{k2}} \cdot \overline{\text{P1}}) ((\text{Ms1} - 2 \text{Ms2}) \text{Ms2} + \overline{\text{P1}} \cdot \overline{\text{P2}})) + \\
& \quad \quad 2 (\overline{\text{k1}} \cdot \overline{\text{P2}}) ((\overline{\text{k2}} \cdot \overline{\text{P1}}) (\overline{\text{P1}} \cdot \overline{\text{P2}} - \text{Ms1 Ms2}) + (\overline{\text{k2}} \cdot \overline{\text{P2}}) (\text{Ms1} (\text{Ms2} - 2 \text{Ms1}) + \overline{\text{P1}} \cdot \overline{\text{P2}}))) + \\
& \quad \frac{1}{(\text{M1} + \text{M2})^2} \text{f2v}^2 (-\overline{\text{k1}} \cdot \overline{\text{k2}}) (\text{Ms1}^2 + \text{Ms2}^2 - 2 (\overline{\text{P1}} \cdot \overline{\text{P2}})) (\text{Ms1 Ms2} - \overline{\text{P1}} \cdot \overline{\text{P2}}) + \\
& \quad \quad 2 (\overline{\text{k1}} \cdot \overline{\text{P2}}) ((\overline{\text{k2}} \cdot \overline{\text{P1}}) (\text{Ms1 Ms2} + \overline{\text{P1}} \cdot \overline{\text{P2}}) + (\overline{\text{k2}} \cdot \overline{\text{P2}}) (\overline{\text{P1}} \cdot \overline{\text{P2}} - \text{Ms1} (2 \text{Ms1} + \text{Ms2}))) + \\
& \quad \quad \left. 2 (\overline{\text{k1}} \cdot \overline{\text{P1}}) ((\overline{\text{k2}} \cdot \overline{\text{P2}}) (\text{Ms1 Ms2} + \overline{\text{P1}} \cdot \overline{\text{P2}}) + (\overline{\text{k2}} \cdot \overline{\text{P1}}) (\overline{\text{P1}} \cdot \overline{\text{P2}} - \text{Ms2} (\text{Ms1} + 2 \text{Ms2}))) \right)
\end{aligned}$$

$s[a_-, b_-] := \text{SP}[a, b];$

$\text{OldT1} = (\text{CA}^2 - \text{CV}^2) \text{Ms}^2 s[k1, k2];$

$\text{OldT2} = (\text{CA} - \text{CV})^2 s[k1, P2] \times s[k2, P1];$

$\text{OldT3} = (\text{CA} + \text{CV})^2 s[k1, P1] \times s[k2, P2];$

$\text{OldT4} = \text{CM Ms} / \text{MN} (2 \text{CA} (s[k1, P1] \times s[k2, P2] - s[k1, P2] \times s[k2, P1]) + \text{CV} (s[k1, k2] (s[P1, P2] - \text{Ms}^2) - (s[k1, P1] - s[k1, P2]) (s[k$

$\text{OldT5} = -\text{CM}^2 / (4 \text{MN}^2) (\text{Ms}^2 s[k1, P2] (3 s[k2, P2] - s[k2, P1]) + \text{Ms}^2 s[k1, P1] (3 s[k2, P1] - s[k2, P2]) + s[k1, k2] (s[P1, P2] -$

$\text{OldT6} = \text{CA}^2 \text{Fq} (2 \text{Ms} + \text{Fq} (\text{Ms}^2 - s[P1, P2])) (s[k1, k2] (\text{Ms}^2 - s[P1, P2]) - (s[k1, P1] - s[k1, P2]) (s[k2, P1] - s[k2, P2]));$

$\text{Eq410ldBracket} = \text{OldT1} + \text{OldT2} + \text{OldT3} + \text{OldT4} + \text{OldT5} + \text{OldT6};$

$\text{Eq410ldFull} = 32 \text{GF}^2 \text{Ccab}^2 \text{Eq410ldBracket};$

$\text{oldLimitRules} = \{\text{Ms1} \rightarrow \text{Ms}, \text{Ms2} \rightarrow \text{Ms}, \text{M1} \rightarrow \text{MN}, \text{M2} \rightarrow \text{MN}, \text{F1V}[_] \rightarrow \text{CV}, \text{F1A}[_] \rightarrow \text{CA}, \text{F2V}[_] \rightarrow -\text{CM}, \text{F2A}[_] \rightarrow 0, \text{F3V}[_] \rightarrow 0, \text{F3A}[_] \rightarrow -(\text{$

$\text{GenOldBracket} = \text{Eq41GeneralBracket} /. \text{oldLimitRules} // \text{Contract} // \text{ExpandScalarProduct} // \text{DiracSimplify} // \text{ComplexExpand} // \text{Factor};$

$\text{GenOldFull} = 32 \text{GF}^2 \text{Ccab}^2 \text{GenOldBracket};$

$\text{CheckOldBracket} = \text{GenOldBracket} - \text{Eq410ldBracket} // \text{Contract} // \text{ExpandScalarProduct} // \text{DiracSimplify} // \text{ComplexExpand} // \text{Factor};$

$\text{CheckOldFull} = \text{GenOldFull} - \text{Eq410ldFull} // \text{Contract} // \text{ExpandScalarProduct} // \text{DiracSimplify} // \text{ComplexExpand} // \text{Factor};$

CheckOldBracket

CheckOldFull

0

0


```

(* ===== *)
(* Quadratic-form decomposition *)
(* ===== *)

ClearAll[
ffList, ffLabels, Fvec, exprQuad, Kmat, KmatReadable,
kinRulesReadable, checkK, independentElements, printElement,
S12, C12, A1, A2, B1, B2
];

ffList = {
f1v, f2v, f3v,
f1a, f2a, f3a
};

ffLabels = {
"f1f1", "f1f2", "f1f3", "f1g1", "f1g2", "f1g3",
"f2f2", "f2f3", "f2g1", "f2g2", "f2g3",
"f3f3", "f3g1", "f3g2", "f3g3",
"g1g1", "g1g2", "g1g3",
"g2g2", "g2g3",
"g3g3"
};

Fvec = ffList;

(* ===== *)
(* Prepare the expression as a polynomial in the form factors *)
(* ===== *)

exprQuad =
Eq41GeneralShort //
Together //
Cancel //
Expand;

(* ===== *)
(* Extract the symmetric matrix K *)
(* ===== *)
(*
(* A = F^T K F *)
(* *)
(* Diagonal terms: *)
(* K_ii = coefficient of F_i^2 *)
(* *)
(* Off-diagonal terms: *)
(* 2 K_ij = coefficient of F_i F_j *)
(* ===== *)

Kmat = Table[
If[
i == j,
Coefficient[exprQuad, ffList[[i]], 2],
1/2 D[exprQuad, ffList[[i]], ffList[[j]]
],
{i, Length[ffList]},
{j, Length[ffList]}
] /. Thread[ffList -> 0];

Kmat =
Kmat //
Map[Together, #, {2}] & //
Map[Cancel, #, {2}] & //
Map[FactorTerms, #, {2}] &;

(* ===== *)
(* Replace FeynCalc scalar products by readable invariants *)
(* ===== *)
(*
(* S12 = P1.P2 *)
(* C12 = k1.k2 *)
(* A1 = k1.P1 *)
(* A2 = k1.P2 *)
(* B1 = k2.P1 *)
(* B2 = k2.P2 *)
(* *)
(* Here k1 is the antineutrino and k2 the charged lepton. *)
(* ===== *)

```

```

kinRulesReadable = {
  Pair[Momentum[P1], Momentum[P2]] => S12,
  Pair[Momentum[P2], Momentum[P1]] => S12,

  Pair[Momentum[k1], Momentum[k2]] => C12,
  Pair[Momentum[k2], Momentum[k1]] => C12,

  Pair[Momentum[k1], Momentum[P1]] => A1,
  Pair[Momentum[P1], Momentum[k1]] => A1,

  Pair[Momentum[k1], Momentum[P2]] => A2,
  Pair[Momentum[P2], Momentum[k1]] => A2,

  Pair[Momentum[k2], Momentum[P1]] => B1,
  Pair[Momentum[P1], Momentum[k2]] => B1,

  Pair[Momentum[k2], Momentum[P2]] => B2,
  Pair[Momentum[P2], Momentum[k2]] => B2
};

KmatReadable =
  Kmat /. kinRulesReadable //
  Map[Together, #, {2}] & //
  Map[Cancel, #, {2}] & //
  Map[FactorTerms, #, {2}] & //
  FullSimplify;

(* ===== *)
(* Exact reconstruction check *)
(* ===== *)
(* This check must use the original Kmat, not KmatReadable, *)
(* because exprQuad still contains FeynCalc Pair objects. *)
(* ===== *)

checkK =
  Together[
    exprQuad - Expand[Fvec . Kmat . Fvec]
  ] //
  Cancel //
  FullSimplify;

Print["Quadratic-form reconstruction check = ", checkK];

(* Expected output: 0 *)

(* Optional symmetry check *)

Print[
  "K-matrix symmetry check = ",
  FullSimplify[Kmat - Transpose[Kmat]]
];

(* Expected output: a 6 x 6 zero matrix *)

(* ===== *)
(* Independent upper-triangular matrix elements *)
(* ===== *)

independentElements = {
  {"Kf1f1", KmatReadable[[1, 1]]},
  {"Kf1f2", KmatReadable[[1, 2]]},
  {"Kf1f3", KmatReadable[[1, 3]]},
  {"Kf1g1", KmatReadable[[1, 4]]},
  {"Kf1g2", KmatReadable[[1, 5]]},
  {"Kf1g3", KmatReadable[[1, 6]]},

  {"Kf2f2", KmatReadable[[2, 2]]},
  {"Kf2f3", KmatReadable[[2, 3]]},
  {"Kf2g1", KmatReadable[[2, 4]]},
  {"Kf2g2", KmatReadable[[2, 5]]},
  {"Kf2g3", KmatReadable[[2, 6]]},

  {"Kf3f3", KmatReadable[[3, 3]]},
  {"Kf3e1", KmatReadable[[3, 4]]}
}

```

```

{ "Kf3g2", KmatReadable[[3, 5]],
{ "Kf3g3", KmatReadable[[3, 6]],

{ "Kg1g1", KmatReadable[[4, 4]],
{ "Kg1g2", KmatReadable[[4, 5]],
{ "Kg1g3", KmatReadable[[4, 6]],

{ "Kg2g2", KmatReadable[[5, 5]],
{ "Kg2g3", KmatReadable[[5, 6]],

{ "Kg3g3", KmatReadable[[6, 6]]
};

(* ===== *)
(* Print each coefficient in readable Mathematica form *)
(* ===== *)

printElement[{name_, expr_}] := Module[{clean},

clean =
  expr //
  Together //
  Cancel //
  FactorTerms //
  FullSimplify;

Print[""];
Print[name, " ="];
Print[InputForm[clean]];
];

Scan[printElement, independentElements];

```

Quadratic-form reconstruction check = 0

$$\text{K-matrix symmetry check} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Kflf1 =

$$A2 \cdot B1 + A1 \cdot B2 - C12 \cdot Ms1 \cdot Ms2$$

Kflf2 =

$$\left(2 \cdot A2 \cdot B2 \cdot Ms1 + 2 \cdot A1 \cdot B1 \cdot Ms2 - A2 \cdot B1 \cdot (Ms1 + Ms2) - A1 \cdot B2 \cdot (Ms1 + Ms2) + C12 \cdot (Ms1 + Ms2) \cdot (Ms1 \cdot Ms2 - S12) \right) / (2 \cdot (M1 + M2))$$

Kflf3 =

$$\left(-2 \cdot A2 \cdot B2 \cdot Ms1 + A2 \cdot B1 \cdot (Ms1 - Ms2) + A1 \cdot B2 \cdot (Ms1 - Ms2) + 2 \cdot A1 \cdot B1 \cdot Ms2 - C12 \cdot (Ms1 - Ms2) \cdot (Ms1 \cdot Ms2 + S12) \right) / (2 \cdot M1)$$

Kflg1 =

$$-(A2 \cdot B1) + A1 \cdot B2$$

$$\text{Kf1g2} =$$

$$-\left(\left(\left(A2*B1 - A1*B2\right)*\left(Ms1 - Ms2\right)\right)/\left(M1 + M2\right)\right)$$

$$\text{Kf1g3} =$$

$$0$$

$$\text{Kf2f2} =$$

$$\begin{aligned} & \left(-2*A1*Ms2*(B1*Ms1 - B2*Ms1 + 2*B1*Ms2) - \right. \\ & \quad C12*(Ms1^2 + Ms2^2 - 2*S12)*(Ms1*Ms2 - S12) + 2*A1*(B1 + B2)*S12 + \\ & \quad \left. 2*A2*(B1*Ms1*Ms2 - B2*Ms1*(2*Ms1 + Ms2) + (B1 + B2)*S12)\right)/(2*(M1 + M2)^2) \end{aligned}$$

$$\text{Kf2f3} =$$

$$\begin{aligned} & \left(A2*B1*(-Ms1^2 + Ms2^2) + A1*B2*(-Ms1^2 + Ms2^2) + \right. \\ & \quad \left. 2*A2*B2*(Ms1^2 - S12) + 2*A1*B1*(-Ms2^2 + S12)\right)/(2*M1*(M1 + M2)) \end{aligned}$$

$$\text{Kf2g1} =$$

$$\left((A2*B1 - A1*B2)*(Ms1 + Ms2)\right)/(M1 + M2)$$

$$\text{Kf2g2} =$$

$$\left((A2*B1 - A1*B2)*(Ms1 - Ms2)*(Ms1 + Ms2)\right)/(M1 + M2)^2$$

$$\text{Kf2g3} =$$

$$0$$

$$\text{Kf3f3} =$$

$$-1/2*\left((-2*(A1 - A2)*(B1 - B2) + C12*(Ms1^2 + Ms2^2 - 2*S12))*(Ms1*Ms2 + S12)\right)/M1^2$$

$$\text{Kf3g1} =$$

$$0$$

$$\text{Kf3g2} =$$

$$0$$

Kf3g3 =

0

Kg1g1 =

A2 * B1 + A1 * B2 + C12 * Ms1 * Ms2

Kg1g2 =

$$\frac{(-2 * A2 * B2 * Ms1 + A2 * B1 * (Ms1 - Ms2) + A1 * B2 * (Ms1 - Ms2) + 2 * A1 * B1 * Ms2 + C12 * (Ms1 - Ms2) * (Ms1 * Ms2 + S12))}{(2 * (M1 + M2))}$$

Kg1g3 =

$$-1/2 * (-2 * A2 * B2 * Ms1 - 2 * A1 * B1 * Ms2 + A2 * B1 * (Ms1 + Ms2) + A1 * B2 * (Ms1 + Ms2) + C12 * (Ms1 + Ms2) * (Ms1 * Ms2 - S12)) / M1$$

Kg2g2 =

$$\frac{(-2 * A2 * Ms1 * (2 * B2 * Ms1 + B1 * Ms2 - B2 * Ms2) + 2 * A2 * (B1 + B2) * S12 + C12 * (Ms1^2 + Ms2^2 - 2 * S12) * (Ms1 * Ms2 + S12) + 2 * A1 * (Ms2 * (B1 * Ms1 - B2 * Ms1 - 2 * B1 * Ms2) + (B1 + B2) * S12))}{(2 * (M1 + M2)^2)}$$

Kg2g3 =

$$\frac{(A2 * B1 * (-Ms1^2 + Ms2^2) + A1 * B2 * (-Ms1^2 + Ms2^2) + 2 * A2 * B2 * (Ms1^2 - S12) + 2 * A1 * B1 * (-Ms2^2 + S12))}{(2 * M1 * (M1 + M2))}$$

Kg3g3 =

$$\frac{((-2 * (A1 - A2) * (B1 - B2) + C12 * (Ms1^2 + Ms2^2 - 2 * S12)) * (Ms1 * Ms2 - S12))}{(2 * M1^2)}$$

file = "/home/kostakis/Wolfram Mathematica/Neutrino_Emission.wl";

txt = Import[file, "Text", CharacterEncoding -> "UTF-8"];

nb = CreateDocument[
{
Cell[txt, "Program",
FontFamily -> "Courier New",
FontSize -> 9,
PageWidth -> WindowWidth
],
},
WindowTitle -> "Mathematica Script",
PageWidth -> 800
];

Export["/home/kostakis/Wolfram Mathematica/Neutrino_Emission_Mathematica.pdf", nb, "PDF"]
"/home/kostakis/Wolfram Mathematica/Erasmus.pdf"

/home/kostakis/Wolfram Mathematica/Neutrino_Emission_Mathematica.pdf

```
/home/kostakis/Wolfram Mathematica/Erasmus.pdf
```

```
Export["/home/kostakis/Wolfram Mathematica/Neutrino_Emission_Mathematica.pdf",  
EvaluationNotebook[], "PDF"]
```

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/home/kostakis/Wolfram Mathematica/Neutrino_Emission_Mathematica.pdf
```

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Export[  
"/home/kostakis/Wolfram Mathematica/Neutrino_Emission_Mathematica.html",  
EvaluationNotebook[],  
"HTML"  
]
```